Particle Swarm optimization

Pseudo code

BEGIN

1. Define Objective function f(x)
2. Initialize parameters
   number of particles (N)
   maximum iteration (T)
   Inertia weight (w)
   Cognitive coefficient (c_1)
   Social coefficient (c_2)
3. initialize particles:
   generate random position and velocities
4. evaluate the fitness of each particle
5. Set each particles personal best (pbest) to its current position
6. Set global best (gbest) to the best personal best among all particles

7. for each iteration from 1 to T DO
   for each particle DO
   → update Velocity
   $V = w * V + c1 * r1 (pbest - X) + c2 * r2 (gbest - X)$
   → update position
   $X = X + V$

evaluate fitness $f(x)$
if current fitness is better than pbest
update pbest = x
end for
end for
8 Display best Solution (gbest) and display its function value $f(gbest)$

Agricultural irrigation Scheduling

problem Statement :-

① minimize water usage while maintaing crops

① objective function f

② to minimize the water usage.

$$F = w_1 \cdot f_{water} + w_2 \cdot f_{stress}$$

② maintain soil moisture within the acceptable range

③ Determine the optimal irrigation amount for each Scheduling period

PSO-Based irrigation Scheduling

Begin
Set N = number of particles
Set T = maximum iterations
Set w, c_1, c_2

for each particle i=1 to N

Randomly initialize irrigation schedule X[i]

Initialize Velocity V[i]

initialize Soil moisture using X[i]

Calculate fitness F[i]

Pbest[i] = X[i]

end for

Gbest = schedule with minimum fitness

for iteration = 1 to T

for each particle i=0 to N

generate random number r1 and r2

$$V[i] = w * V[i] + c1 * r1 * (Pbest[i] - X[i]) + c2 * r2 * (Gbest - X[i])$$

$$X[i] = [X[i] + V[i]]$$

Restrict irrigation amount

$$0 <= X[i][t] <= Imax$$

calculate fitness

F[i] = total irrigation water + constraint penalties

~~if F[i] <~~

end for

End for

print optimal values

→ optimal 5-day irrigation schedule

Day 01: planned irrigation = 15.00mm | forecasted rain = 0.0mm
| crop use (ETc) = 3.57mm

Day 02: planned irrigation = 6.36mm | forecasted Rain = 0.0mm
| Crop use (ETc) = 3.82 mm

Day 03: planned irrigation = 6.27mm | forecasted Rain
= 0.00mm | Crop use (ETc) = 4.08 mm

Day 04: planned irrigation = 0.00mm | forecasted Rain
= 0.00mm | Crop use (ETc) = 4.25 mm

Day 05: planned irrigation = 0.0mm | forecasted rain
= 12.mm | crop use (ETc) = 3.23 mm

total Applied irrigation water = 27.62 mm

O/P Seen
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